

T11 - Span approach: data check and decision

EmpowerLens | Module: Span selection | Source: span_check.py on Annotated_data.csv (2,530 rows)

1. What was and was not run

span_check.py did not train or run the three approaches. It measured properties of the data: where each 'Distorted part' span sits inside its 'Patient Question', how it aligns with sentence boundaries, and what trivial sentence-picking baselines score. The comparison in section 3 is therefore **data-implied fit**, reasoned from those measurements. No model has been fitted for sentence classification, BIO tagging or extractive QA; the first real model result comes in T13.

2. Numbers (distorted rows only, span_check.py v2 on the full Annotated_data.csv)

Measure	Value
Distorted rows / with a span	1,597 / 1,597 (100%). No-distortion rows with a span: 0 of 933 (36.9% of the corpus)
How spans were located	exact 1,355 (84.8%) normalized 27 (1.7%) anchor (first/last 6 words) 199 (12.5%) fuzzy 3 (0.2%) unlocated 13 (0.8%)
Anchored rows: region vs quote	region is a median 2.07x the quoted span; 73.9% are >1.5x. Only 37.8% of sentences inside the region appear in the quote (all of them in 3 of 199 rows)
Sentences per text (located, n=1,584)	mean 10.69, median 8, p90 21, max 83; single-sentence texts 19 (1.2%); >=3 sentences 95.7%; words/text median 159, p90 382
Span vs sentences	fragment of one sentence 89 (5.6%) one whole sentence 885 (55.9%) several sentences 610 (38.5%)
Gold sentences per row	1: 974 (61.5%) 2: 363 (22.9%) 3+: 247 (15.6%). Rule: overlap for exact/normalized/fuzzy rows, containment for anchored rows
Non-adjacent gold sentences	182 rows (11.5% of located; 29.8% of rows with 2+ gold). All 182 are anchored rows (182 of 199 anchored)
Labelling-rule sensitivity	'any overlap' vs '>=20% of sentence' disagree on 15 non-anchored rows (1.1%)
Multi-fragment spans	42 (2.6%) contain an ellipsis or newline
Span size	median 29 words, mean 36.5, p90 67; median share of text 0.20; spans >=90% of text: 60 (3.8%)
Trivial selector baselines (hit = overlaps gold)	first 0.154 last 0.248 random 0.242 (texts with 2+ sentences, n=1,565: 0.144 / 0.239 / 0.233)
Selection-bias check	exact+normalized rows: median 27 words, 30.2% multi-sentence anchor+fuzzy rows: median 53 words, 95.0% multi-sentence
Several-sentence rate by secondary label	no secondary 28.2% with secondary 68.2%
Several-sentence rate by class	highest: all_or_nothing 54.5%, magnification 51.3% lowest: mind_reading 24.7%, should_statements 28.0%

3. Data-implied fit of the three approaches

Approach	What the data suggest	Fit
Sentence classification	Only 5.6% of spans are a fragment of a single sentence; the rest cover one whole sentence (55.9%) or several (38.5%). The labelling rule barely matters (1.1%). Median 8 sentences per text, so selection is a real task; baselines to beat are about 0.24 (last/random). 38.5% of rows have 2+ gold sentences, so score each sentence independently (sigmoid), not pick-one.	Best

Approach	What the data suggest	Fit
BIO token tagging	Spans are long (median 29 words), so one wrong tag fragments the output and needs reassembly rules. Few spans are sub-sentence, so token precision adds little. 12.5% of quotes were edited or joined and had to be anchor-matched, so exact character offsets do not exist for them (0.8% could not be located at all).	High cost, small gain
Extractive QA	Assumes one contiguous span, but 11.5% of located rows have non-adjacent gold sentences (the annotators joined separate sentences), and 42 more quotes contain an ellipsis or newline. One span is stored per row, not per distortion type, and rows with a secondary distortion have 2+ gold sentences 68.2% of the time. The 933 no-distortion rows have no span, so QA needs null-answer handling (SQuAD 2.0 style) or a detector gate.	Poor

4. Decision paragraph (report text)

We frame span selection as sentence-level multi-label classification. In the annotated corpus, only 5.6% of located spans are a fragment of a single sentence; the rest cover one whole sentence (55.9%) or several (38.5%), so the sentence is the natural unit of supervision, and labelling a sentence positive when the span overlaps it at all, rather than at 20% coverage, changes the label count on only 1.1% of rows. Texts are long (median 8 sentences, 95.7% with at least three), so selection is non-trivial, and trivial baselines (first, last, random sentence) hit only 0.15 to 0.25. A row has two or more gold sentences in 38.5% of cases, and in 11.5% of rows those sentences are not adjacent because annotators joined separate quotations, so the selector scores each sentence independently rather than choosing one. BIO token tagging would require exact character offsets that do not exist for the 12.5% of quotes that were edited or joined, and reassembly of spans that are long (median 29 words). Extractive question answering assumes one contiguous span, which fails for the non-adjacent cases and for the 36.9% of rows with no distortion and therefore no span. Sentence selection also supports a select-then-classify pipeline (Jain et al., 2020) whose selector can be evaluated separately from the classifier.

5. Proposed T12 labelling rule and caveats

Rule. A sentence is positive if it overlaps the located span (exact, normalized, fuzzy rows). For anchored rows (quotes that were edited or joined), a sentence is positive only if its full text appears inside the quote; labelling everything between the anchors would over-label (region sentences contained in the quote: 37.8%). Exclude the 13 unlocated rows from selector training and evaluation, or hand-label them: ids 160, 306, 315, 405, 436, 887, 1536, 1655, 1697, 1702, 1713, 2445, 4523.

Caveats. (1) 'Several sentences' means the span touches 2+ sentences; whether those spans start or end mid-sentence at their edges was not measured. (2) Containment can under-label a sentence that the annotator only partly quoted. (3) The regex splitter over-splits informal text: for example it cut after 'appt.' and after '.' before a lowercase word, so sentence counts and baselines are slightly inflated. The 1.1% rule disagreement suggests labels are robust to this, but PySBD (Sadvilkar and Neumann, 2020) is a drop-in check before T12 is final. (4) 'Annotators flagged sentences' comes from the project brief's reading of the dataset paper and was not verified against the paper here. (5) The BIO and QA verdicts are reasoned from data, not measured; the first model result comes in T13.

6. References

Verified (via ACL Anthology / arXiv): DeYoung et al. (2020), ERASER: A Benchmark to Evaluate Rationalized NLP Models, ACL 2020, 2020.acl-main.408. Read, Dridan, Oepen and Solberg (2012), Sentence Boundary Detection: A Long Solved Problem?, COLING 2012 Posters, pp. 985-994. Sadvilkar and Neumann (2020), PySBD: Pragmatic Sentence Boundary Disambiguation, NLP-OSS 2020, 2020.nlpss-1.15.

From memory, confirm before citing: Jain, Wiegrefe, Pinter and Wallace (2020), Learning to Faithfully Rationalize by Construction, ACL 2020. Lei, Barzilay and Jaakkola (2016), Rationalizing Neural Predictions, EMNLP 2016. Rajpurkar et al. (2016), SQuAD, EMNLP 2016. Rajpurkar, Jia and Liang (2018), Know What You

Don't Know: Unanswerable Questions for SQuAD, ACL 2018. Ramshaw and Marcus (1995), Text Chunking using Transformation-Based Learning.